

Tutorial 4



fig. (1)

Ex.1: **Mosaic-based privacy protection for the face.** Please replace the face region of `tst1.jpg` with the corresponding mosaic, as shown in fig. (1).

Hints:

- You could use the face detector in the previous tutorial PPT to find the face region.
- The mosaic could be obtained by patch down-sampling and then nearest up-sampling.
(`cv2.resize(...)`)



fig. (2)

Ex.2: Use 2D filtering operation to **sharpen and add motion blur** for `tst2.jpg` **separately**.

Hints:

- You could define the sharpening kernel according to the lecture PPT 1-2 using `np.array()`.
- You could use `cv2.filter2D(...)` to conduct the filtering.
- Motion blur could be regarded as a kind of blur that only happened in one direction.
(e.g., horizontal)

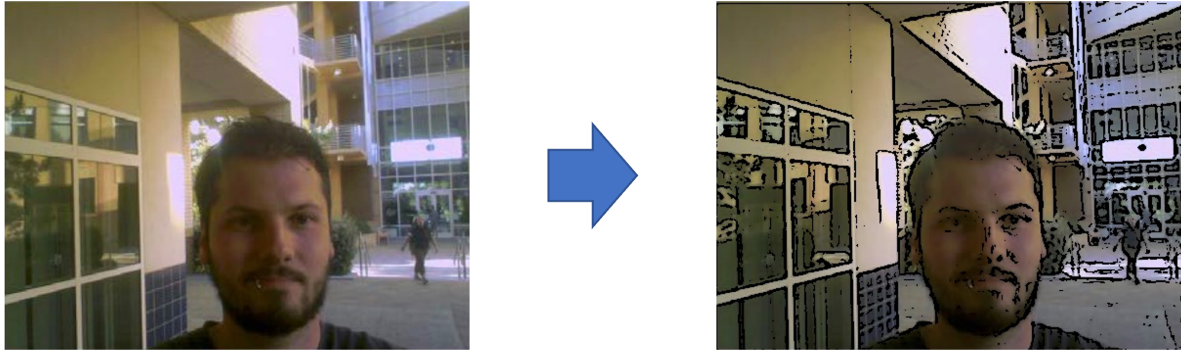


fig. (3)

Ex.3: **Cartoonizing an Image.**

Hints:

- a. The Laplacian filtering and thresholding operations could be used to obtain pencil-like outline. (`cv2.Laplacian(...)` and `cv2.threshold(...)`)
- b. The bilateral filter could be used to smoothen the rough areas in the image and keep the edges intact. (`cv2.bilateralFilter(...)`)
- c. The final result could be a combination of the thick boundary and the smoothened colored areas.
- d. Preprocessing could reduce the noise in pencil-like outlines.